A systems biology approach reveals a lncRNA-miRNA-mRNA regulatory network in Zika virus-infected macrophages that were differentiated with vitamin D

Julieta M Ramírez-Mejía[a], Geysson Javier Fernandez[b,c], Silvio Urcuqui-Inchima[b*]

[a]Instituto Nacional de Cancerología, Bogotá, Colombia

[b]Grupo Inmunovirología. Departamento de Microbiología y Parasitología, Universidad de Antioquia UdeA, Medellín, Colombia.

[c]Grupo Biología y Control de Enfermedades Infecciosas, Universidad de Antioquia UdeA, Medellín, Colombia.

jramirez@cancer.gov.co

UNIVERSIDAD DE ANTIOQUIA

BCEI Biología y Control de Enfermedades Infecciosas

Introduction

Zika virus (ZIKV) infection poses significant global health risks, particularly due to its links to congenital malformations and severe neurological issues. Recent research has enhanced understanding of immune responses to ZIKV, revealing the importance of innate immunity and the immunomodulatory effects of Vitamin D (VitD) on macrophages. This study investigates the interplay between VitD, non-coding RNAs (miRNAs and lncRNAs), and mRNA in ZIKV-infected macrophages, utilizing RNA-Seq and miRNA-Seq data to construct a regulatory network and explore molecular interactions.

Objectives

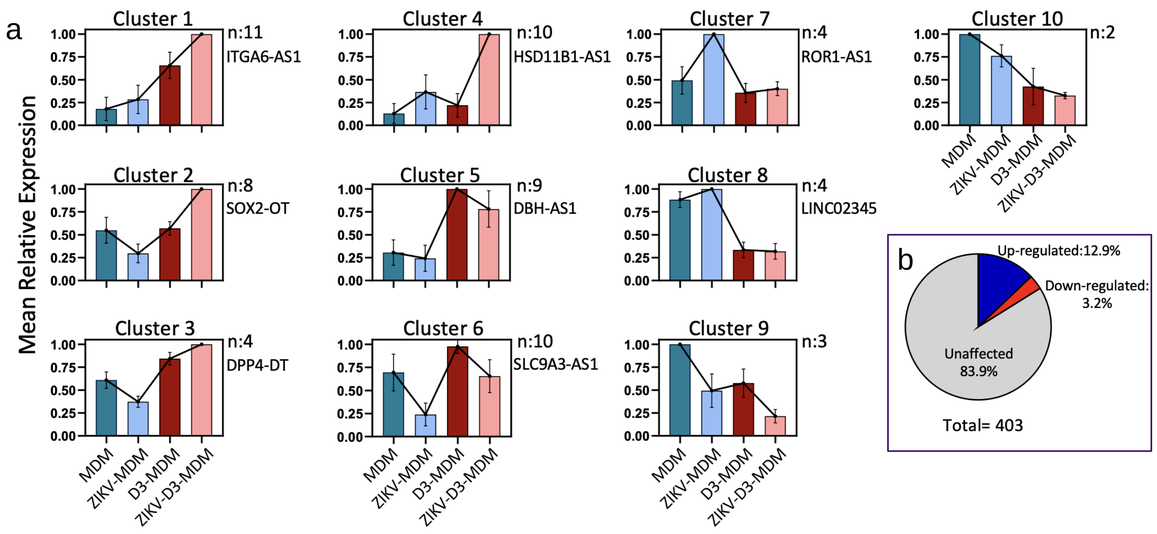
- Extract expression profiles of mRNA, miRNA, and lncRNA in ZIKV-infected monocyte-derived macrophages (MDMs) differentiated with or without Vitamin D.
- To construct a competing endogenous RNA (ceRNA) regulatory network for analyzing the interactions between Vitamin D, microRNAs (miRNAs), and long non-coding RNAs (lncRNAs) in Zika virus (ZIKV)-infected MDMs.
- Explore the specific regulatory pathways and networks involving non-coding RNAs and mRNA in the context of Vitamin D-mediated immune modulation during ZIKV infection.

Methods

- Human monocytes were cultured in RPMI-1640 and differentiated into MDMs with or without 1 nM Vitamin D3, with medium refreshed daily until day 6.
- On day 6, MDMs and D3-MDMs were infected with ZIKV strain COL345Si-Asian at a multiplicity of infection (MOI) of 5 in serum-free RPMI-1640.
- Total RNA was extracted, miRNA libraries were prepared from infected and uninfected samples using a single adapter method, sequenced as 50 bp single-end reads with 7 to 12 million reads per sample.
- Read alignment was performed for miRNAs, lncRNAs, and mRNAs, followed by differential expression analysis.
- VDR motif analysis, gene enrichment analysis, miRNA target prediction, and construction of a ceRNA network were conducted, along with statistical analyses.

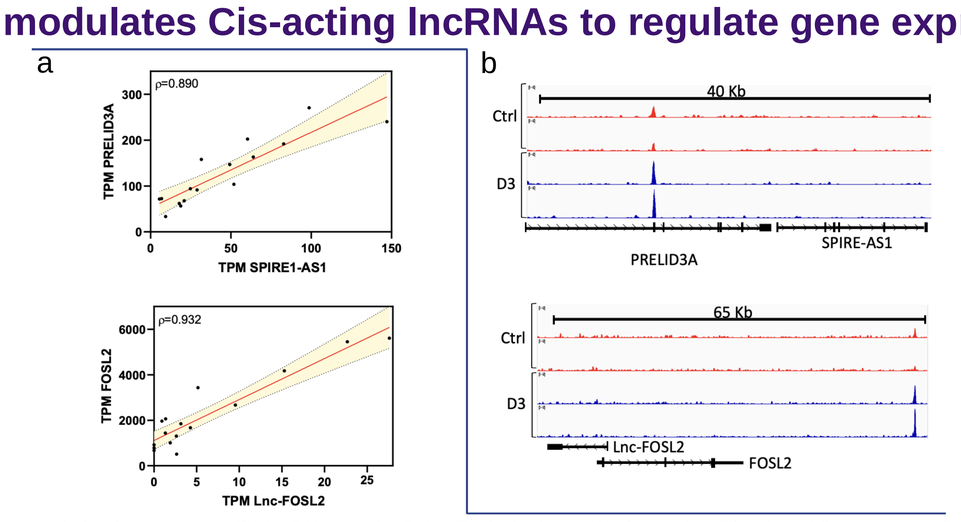
Results

1. VitD-mediated modulation of lncRNA in ZIKV-infected MDMs



1(a) Up-regulated and **(b)** Down-regulated genes were categorized into six and four clusters, respectively, using k-means clustering based on their expression patterns across the four experimental groups. "n" indicates the number of lncRNAs **(c)** Pie chart illustrating the up-regulated and down-regulated lncRNA responses to Vitamin D in ZIKV-infected MDMs.

2. VitD modulates Cis-acting lncRNAs to regulate gene expression



2. (**a**) Pearson correlation between specific lncRNAs and mRNA. Each dot is a pair of samples. The dotted line presents the error. Pearson correlation value is shown inset. **(b)** Genomic tracks show VDR ChIP-seq data, illustrating VDR enrichment profiles in THP1 control macrophages (red) and vitamin D-treated cells (blue).

Acknowledgments

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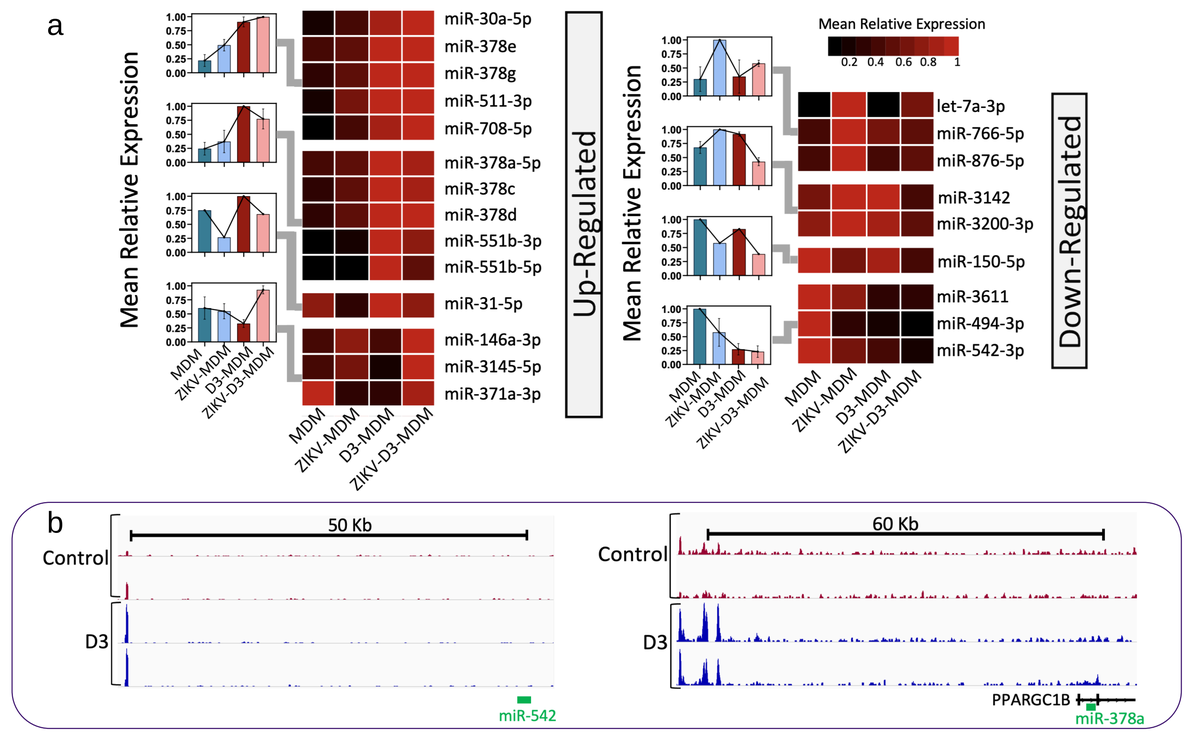
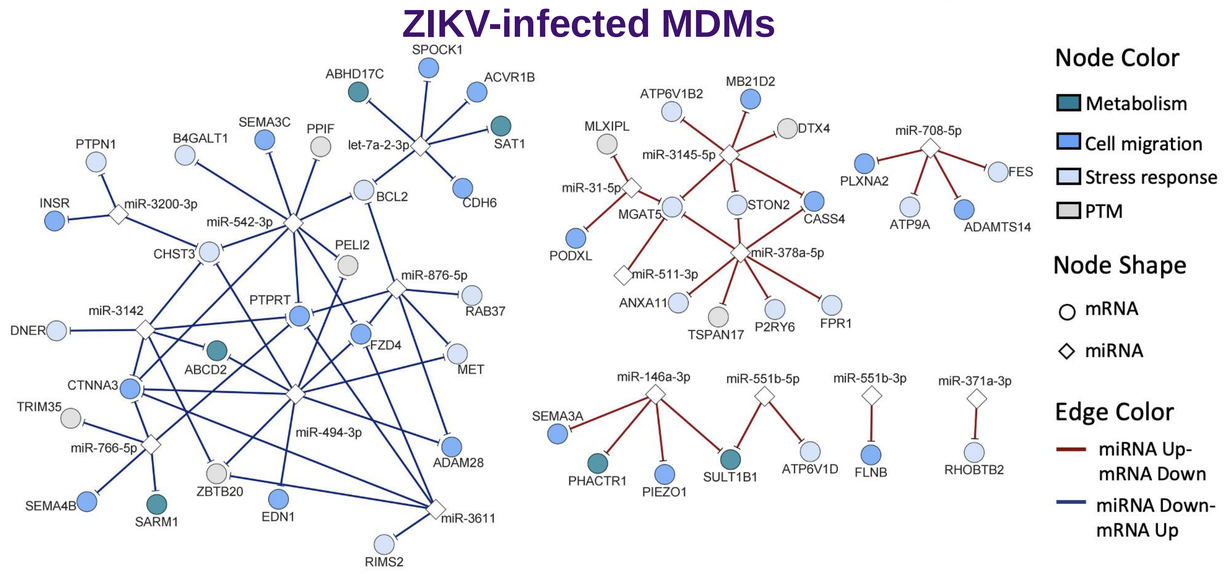
3. VitD modulates miRNA expression in ZIKV-infected MDMs



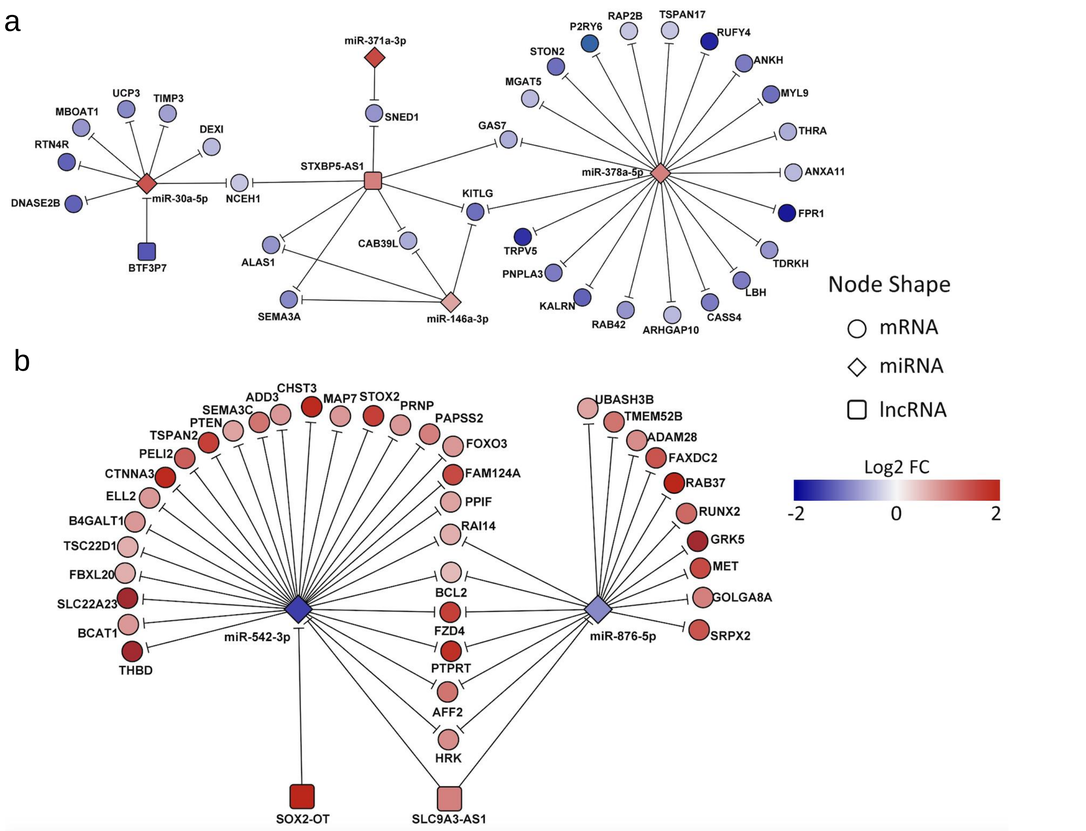
3. **(a)** Heatmap of differentially expressed miRNAs divided into eight clusters, using k-means, based on their pattern of transcript levels in the four samples. **(b)** Genomic tracks illustrate VDR ChIP-seq data in THP1 control macrophages (red) and vitamin D-treated macrophages (blue), covering a region up to 60 kb around the TSS of the miRNAs.

4. VitD-regulated miRNAs modulate key biological processes in ZIKV-infected MDMs



4. Interactome between mRNA and the respective potential regulatory microRNAs. Circular nodes represent mRNAs, diamond-shaped nodes represent miRNAs. Node color represents the biological process, and the edge color represents the up and down interactions between miRNA and mRNA.

5. Regulation by VitD via ceRNA Mechanisms in ZIKV-Infected MDMs



5. Interactome of potential cross-regulations among mRNAs, miRNAs, and lncRNAs for both **(a)** down-regulated and **(b)** up-regulated genes. Circular nodes represent mRNAs, diamond-shaped nodes represent miRNAs, and square nodes represent lncRNAs.

Conclusions

Our findings suggest:

1.The pivotal involvement of ceRNA regulation in various diseases and cellular processes, emphasizing its relevance in ZIKV pathology.

2.The intricate interplay between miRNAs and lncRNAs, influenced by vitamin D, that shapes host-virus interactions in macrophages.

3.The necessity for further experimental validation to confirm the identified regulatory relationships and their functional significance.

4.The potential for these insights to inform therapeutic strategies targeting miRNA and lncRNA pathways in viral infections and immune-related disorders.